



Tata Tech Innovent

KAVACH

Connected & Self-Defending Edge Intelligence

for Industrial Systems

Theme Focus

Edge AI for Connected, Secure & Intelligent Industrial Systems



Watches

Continuously monitors machine health.



Explains

Communicates problems clearly in plain language.



Defends

Protects itself even when network is offline.



Team Kavach

KIIT University

Team Members



Koustubh Verma

Hardware & IoT Architecture



Gyan Prakash

AI/ML & Edge Intelligence



Sunnik Chatterjee

Backend & Security Systems



Pratik Halder

UI/UX & Product Design

Team Kavach

Meet Our Builders

The multidisciplinary engineering team behind KAVACH Edge Intelligence



Gyan Prakash

AI/ML Engineer

Key Focus

Edge AI deployment & real-time anomaly detection models.

Past Project

Built a tinyML model running on low-power ARM microcontrollers.



Koustubh Verma

Embedded Systems

Key Focus

Hardware design, sensor arrays & industrial IoT architecture.

Past Project

Designed secure, ruggedized telemetry modules for factory floors.



Sunnik Chatterjee

Data Engineer

Key Focus

Distributed storage, pipeline security & local edge databases.

Past Project

Optimized a high-throughput pipeline handling 10k messages/sec.



Pratik Halder

Full Stack Developer

Key Focus

Responsive operator dashboards & secure API endpoints.

Past Project

Developed real-time visual grid to control remote factory nodes.

The Core Challenge

Factories are running blind, wasteful, and unprotected.



Unplanned Downtime Bleeds Cash

Downtime events cost manufacturers thousands of dollars per single hour of failure, impacting quarterly margins instantly.



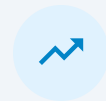
Knowledge Loss & Delayed Triage

Tracing failure root causes takes hours and relies entirely on tribal knowledge from senior engineers who eventually retire or leave.



Invisible Semantic Cyberattacks

Malicious code can manipulate machine parameters while spoofing the network to report normal operation—a threat almost no commercial tools detect.



Market Urgency

Multi-Billion

The predictive maintenance market is surging toward tens of billions of dollars by the early 2030s, fueled by strong industrial demand and market pull.

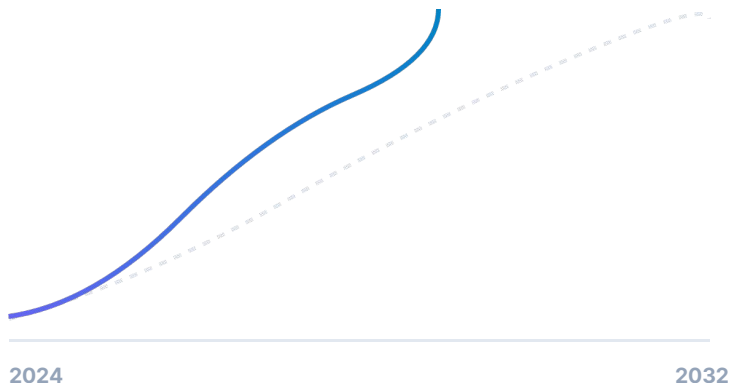
Market Opportunity

High CAGR expansion driven by underserved mid-sized segments

Projected Growth Through 2032

Double-Digit CAGR

The AI-driven predictive maintenance market experiences unprecedented compounding acceleration fueled by massive industrial intelligence adoption.



Manufacturing Plants

Unlocking uptime metrics across production lines.



Construction Sites

Real-time telemetry tracking on critical assets.



Heavy Machinery Fleets

Intelligent diagnostic modeling for distributed networks.

The Underserved Segment

Most existing solutions are enterprise platforms out of reach for mid-sized plants, leaving a large underserved segment.

Our Goal

"Build an **edge-first system** that detects problems early, explains their root cause in plain language, recommends the energy-efficient fix, and protects machines from attacks the network cannot see, **all without depending on the cloud** ."



Detect



Explain



Optimize



Protect

Edge-to-Server Intelligence Pipeline



STEP 01

Continuous Sensing

Sensors installed on each industrial machine collect multi-modal telemetry:

- Vibration & Temp
- Pressure & Current



STEP 02

Edge AI Filtering

Local edge devices execute micro-models to isolate anomalies instantly:

- Real-time detection
- Smart event filtering



STEP 03

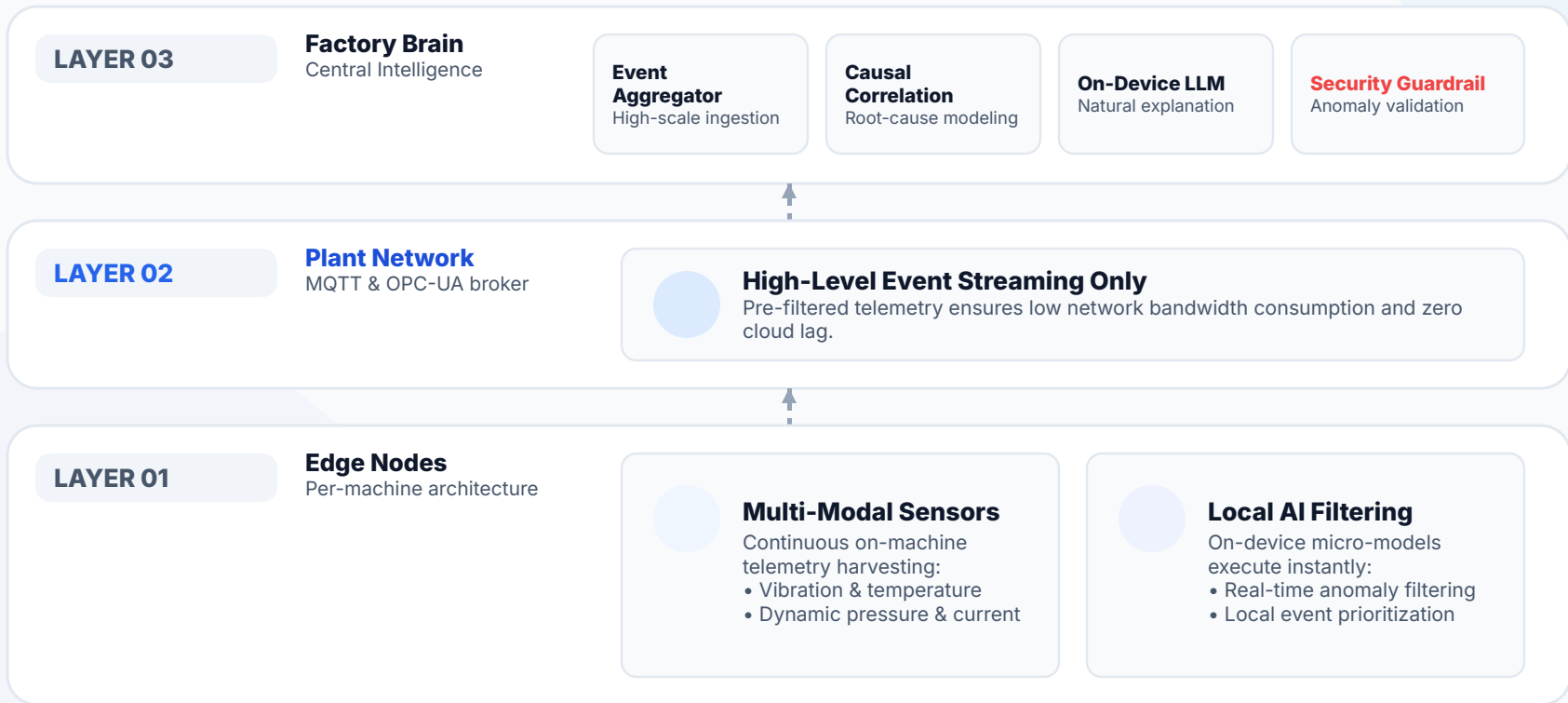
Factory Brain

Central server correlates data and prepares actions for approval:

- LLM root-cause analysis
- Human-in-the-loop audit

SYSTEM ARCHITECTURE

Edge-to-Server Intelligence Infrastructure



Beyond Alarms: Explaining Why, Not Just What

BEFORE: TRADITIONAL ALARMS



Vibration Spike Detected

Sensor VIB-049 exceeds critical threshold (8.2 mm/s)



High Temperature Warning

Bearing housing temp at 92°C (Trend: Rising)



Hydraulic Pressure Drop

System line pressure fell below 180 bar



AFTER: EXPLAINABLE EDGE AI

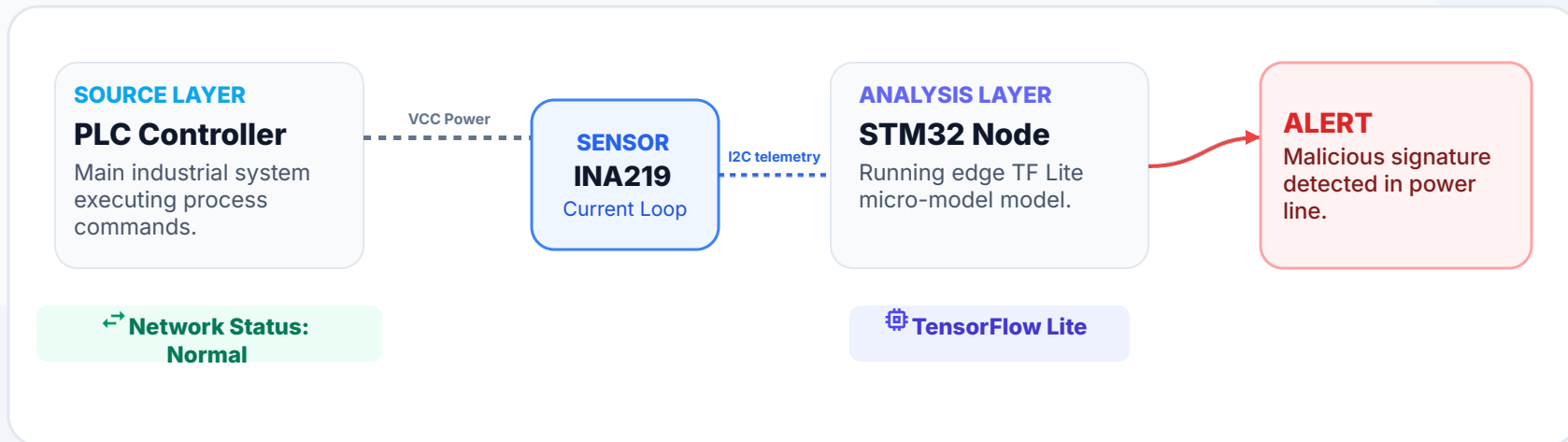


ROOT-CAUSE INSIGHT

"Bearing wear on **Machine A** is increasing motor load, raising energy use by **15 percent**."

THREAT DETECTION

Catching Attacks the Network Cannot See



"Detects **malicious code execution** through power signature analysis, operating **fully independent** of the network layer."

OPTIMIZATION BENEFIT

Turning Waste Into Savings

BEFORE

Engine Speed

2,000 RPM

Fuel Consumption

15 L/hr



AFTER EDGE AI

Engine Speed

900 RPM

Fuel Consumption

5 L/hr

EFFICIENCY METRIC

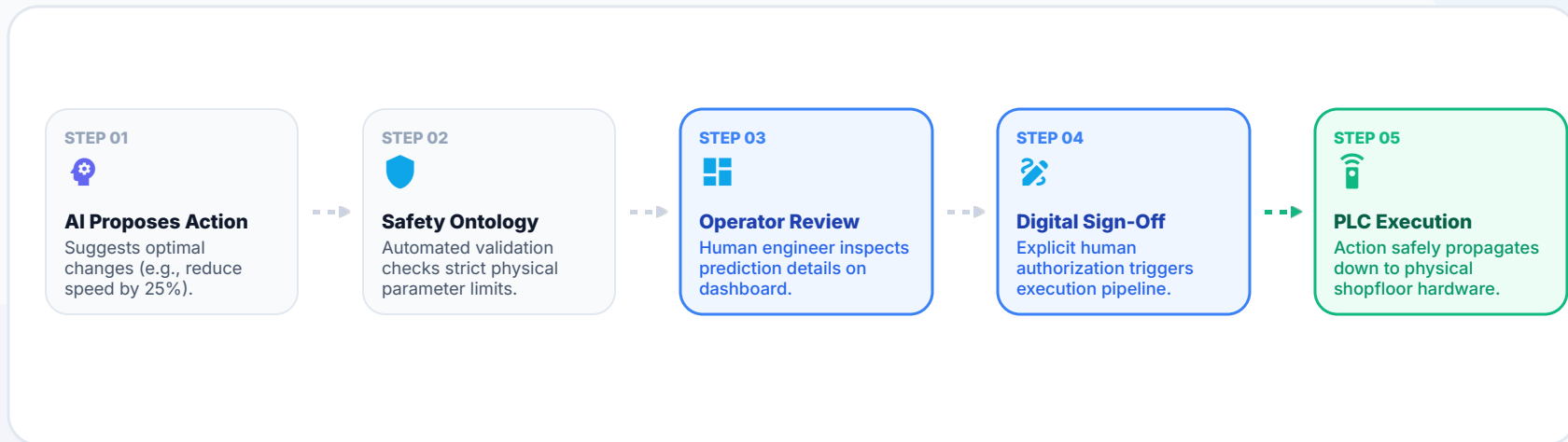


78
EFFICIENCY
SCORE



"Identifies waste like **high RPM at low load** and recommends **specific, quantified fixes** to lock in savings."

AI Recommends, Humans Approve



"Zero-bypass architecture guarantees **no AI recommendation ever touches real machinery** without explicit, authenticated human operator sign-off."

TECHNICAL RESOLUTIONS

Challenges We Tackled

CHALLENGES ENCOUNTERED

CHALLENGE 01

Faulty Sensors vs. Real Hazards

Distinguishing sensor malfunctions and drift from actual, critical process abnormalities.

CHALLENGE 02

Offline Performance Constraints

Keeping processing speeds high and latency low in environments without reliable internet access.

CHALLENGE 03

Unsafe Autonomous Actions

Preventing edge-running AI loops from autonomously initiating dangerous physical movements on machinery.

ENGINEERED SOLUTIONS

SOLUTION

Cross-Channel Autoencoders

Implemented real-time cross-channel consistency checks driven by an unsupervised autoencoder model.

SOLUTION

Edge-Filtered MQTT Events

Sending only high-level compiled events over lightweight MQTT instead of continuous raw data streams.

SOLUTION

Ontology Checks & Sign-Off

Enforced physical parameter semantic ontology limits paired with cryptographic human validation.

TECHNOLOGY STACK

Industrial Edge Intelligence Stack



HARDWARE LAYER

Industrial Edge Devices & Sensors

Raspberry Pi, Jetson Nano, STM32 microcontrollers, INA219 current sensors, and industrial temperature, vibration, and pressure telemetry.



EDGE AI LAYER

Embedded ML & DSP Models

Optimized inference pipelines using TensorFlow Lite, TF Lite Micro, Random Forest, CNN architectures, and unsupervised Autoencoders.



REASONING LAYER

Local Cognitive & Causal Logic

On-device quantized SLMs (Phi-3 mini, Gemma) coupled with Bayesian causal reasoning networks for explainable fault diagnostics.



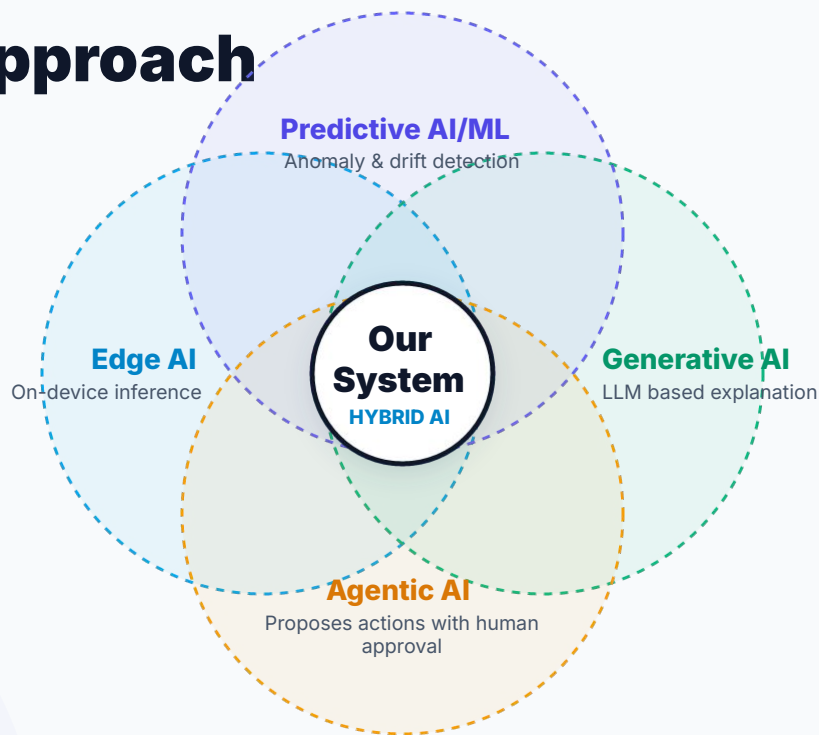
CONNECTIVITY & BACKEND

Data Pipelines & Operations

MQTT, OPC-UA, Node.js/FastAPI, React with Tailwind CSS, and optimized edge/cloud databases (SQLite & InfluxDB time-series).

INTELLIGENCE ARCHITECTURE

A Hybrid AI Approach



Our unified architecture synthesizes **real-time efficiency, predictive diagnostics, explainable insights,** and **safe control loops** .

PERFORMANCE IMPACT

Key Results & Expected Outcomes



20 hrs

Early warning before failure events occur.



~65%

Fuel reduction achieved during key idle periods.



Mins

Root cause time cut down drastically from hours.



~0

Security gap closed where commercial coverage fails.

These quantitative milestones demonstrate how our architecture enhances **operational reliability, system security, and environmental efficiency**.

FUTURE ROADMAP

What's Next



PHASE 1
Multisensory



PHASE 2
Federated AI



PHASE 3
Edge 5G



PHASE 4
Verification

Acoustic & Thermal

Add acoustic and thermal fusion sensing for specialized use cases like steel plants.

Cross-Site Learning

Deploy federated learning across multiple factory sites to handle different wear patterns.

Network Handover

Integrate 5G network handover intelligence for mobile robots and AGVs.

Formal Safety

Full formal verification of LLM generated control commands for safe autonomous operation.

Our strategic expansion pathways drive the platform toward [multisensory industrial telemetry](#), [collaborative intelligence](#), and [guaranteed autonomous safety](#).



PROJECT TIMELINE

Implementation Roadmap

PHASE	TIMELINE	DELIVERABLES & OBJECTIVES
 Phase 1	Weeks 1-2	Sensor Setup & Edge Anomaly Detection Deploy core physical sensors and configure real-time edge anomaly detection algorithms on 1-2 target machines.
 Phase 2	Weeks 3-4	MQTT Integration & Factory Brain Establish unified MQTT messaging pipelines to stream live sensor telemetry directly into the Factory Brain dashboard.
 Phase 3	Weeks 5-6	Local LLM Integration Deploy and fine-tune a secure local Large Language Model for instant, natural language root cause explanations of failures.
 Phase 4	Weeks 7-8	Security Watchdog & Human Sign-Off Build the cybersecurity intrusion detection loop alongside an intuitive operator UI for manual human-in-the-loop approval.
 Phase 5	Weeks 9-10	Full System Integration & Demo Prep Execute end-to-end integration testing across all architecture layers and polish the interface for live demonstration.

Our structured 10-week execution plan scales from **on-device hardware deployment** to a **secure, fully-integrated AI system**.

PROJECT CLOSING

Thank You

We appreciate your time and consideration.



DEVELOPMENT TEAM
Team LastMinuteHustlers



EMAIL ADDRESS
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PROJECT RESOURCES
github.com/koustubh-v/Kavach